

hw_4_building_functions

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Defining the contract

Inputs:

- *ncount* - total number of individuals of each species (these inputs could be multiple values)
- N -the total number of individuals of all species (this input is a single value calculated within the function)

Output: D - Simpson's Diversity Indices, ranges between 0 and 1.

- Higher values indicate lower diversity (with this version of the equation)
- 0 represents infinite diversity
- 1 indicates no diversity

Parameters: *ncount*, each number indicating the number of individuals observed for one species. Multiple values represent multiple species.

- i.e., `simpsons_index(c(15,20,30))` represents 3 species with 15, 20, and 30 individuals each

Implement the function in R

```
#' Simpson's Diversity Index
#'
#' This function measures the probability that two individuals randomly selected from a sample
#' @param ncount the number of individuals of a particular species (quantity)
#' @return Simpsons Index (D)

# function definition in-line
simpsons_index = function(ncount) {
  N = sum(ncount)
  D = sum((ncount/N)^2)
```

```
return(D)
}
```

Testing the Function

```
# Test 1: Single species
simpsons_index(c(20))
```

```
[1] 1
```

```
## expected output D = 1 (no diversity)
```

```
# Test 2: Two species, equally distributed
simpsons_index(c(20,20))
```

```
[1] 0.5
```

```
## expected output is a moderate D (0.5)
```

```
# Test 3: Randomly sampled data with 5 species
set.seed(100)
species_counts <- sample(1:100, size = 5, replace = TRUE)
simpsons_index(species_counts)
```

```
[1] 0.2237224
```

```
## expected output - varies between sample data
```